

Bank Client Churn

Classification Competition

Machine Learning Model Comparison & Hyperparameter Tuning

Dataset: 10,127 clients | 19 features | Target: Status (Churn / Existing)

Attrited: 1,627 (16.1%) | Existing: 8,500 (83.9%)

Algorithms: Logistic Regression · Decision Tree · Random Forest · Gradient Boosting · XGBoost · LightGBM · KNN · SVM

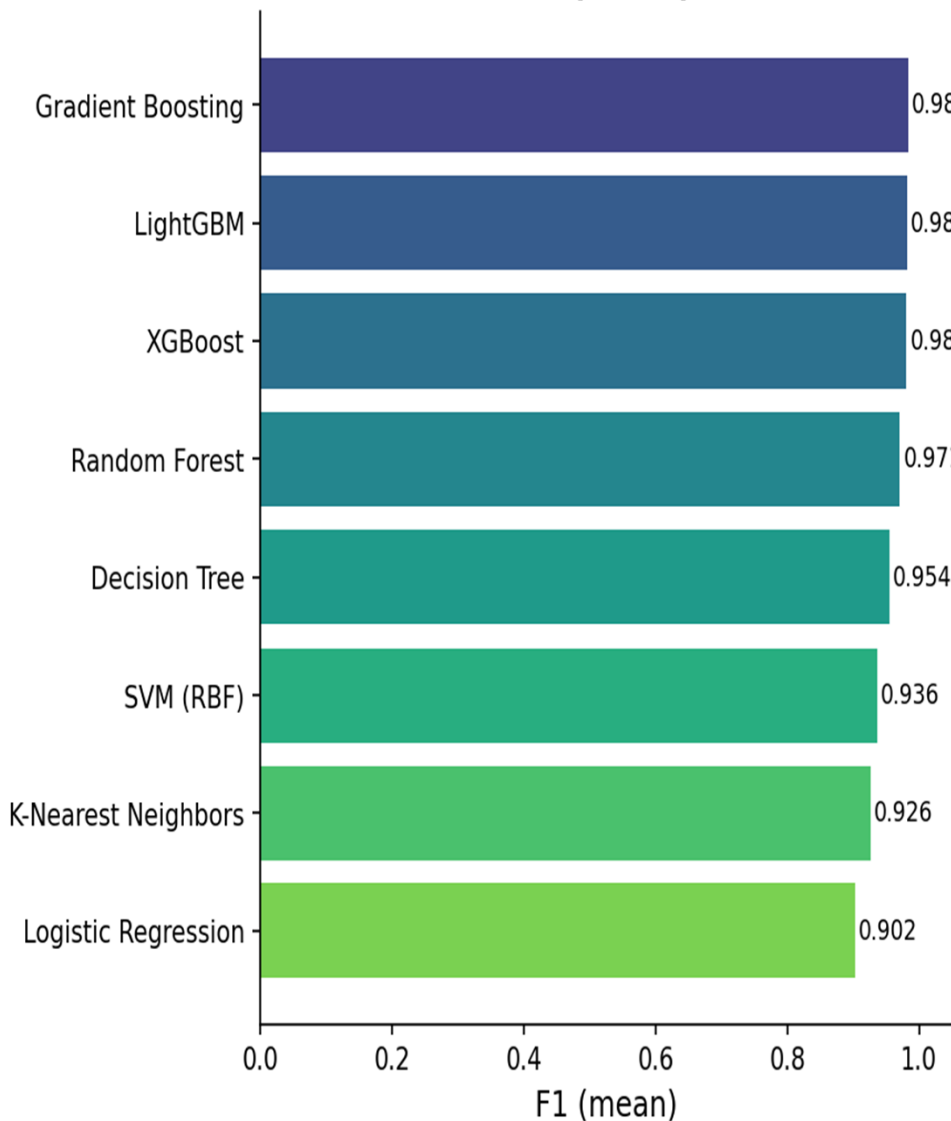
Evaluation: 5-Fold Stratified Cross-Validation



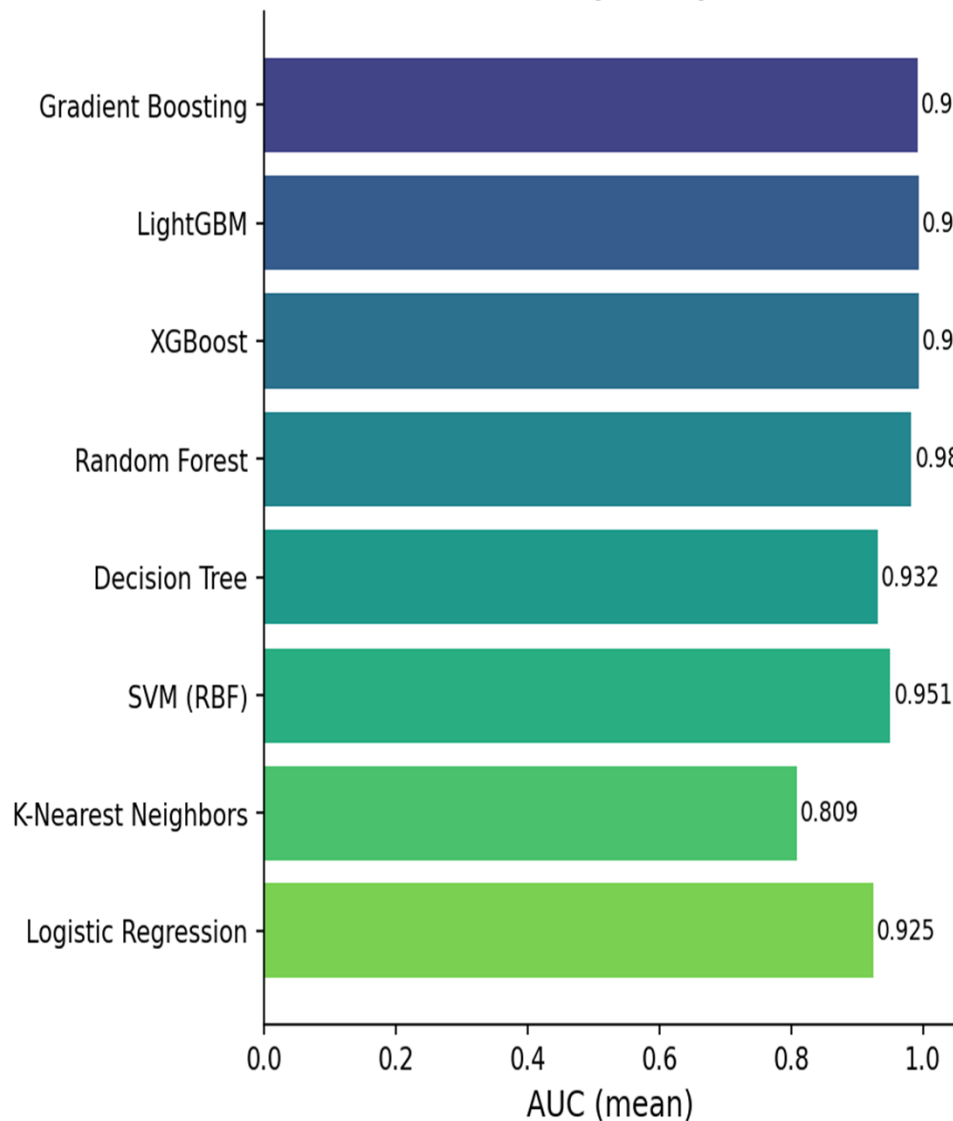
Model Leaderboard

Bank Churn Classification — Model Comparison

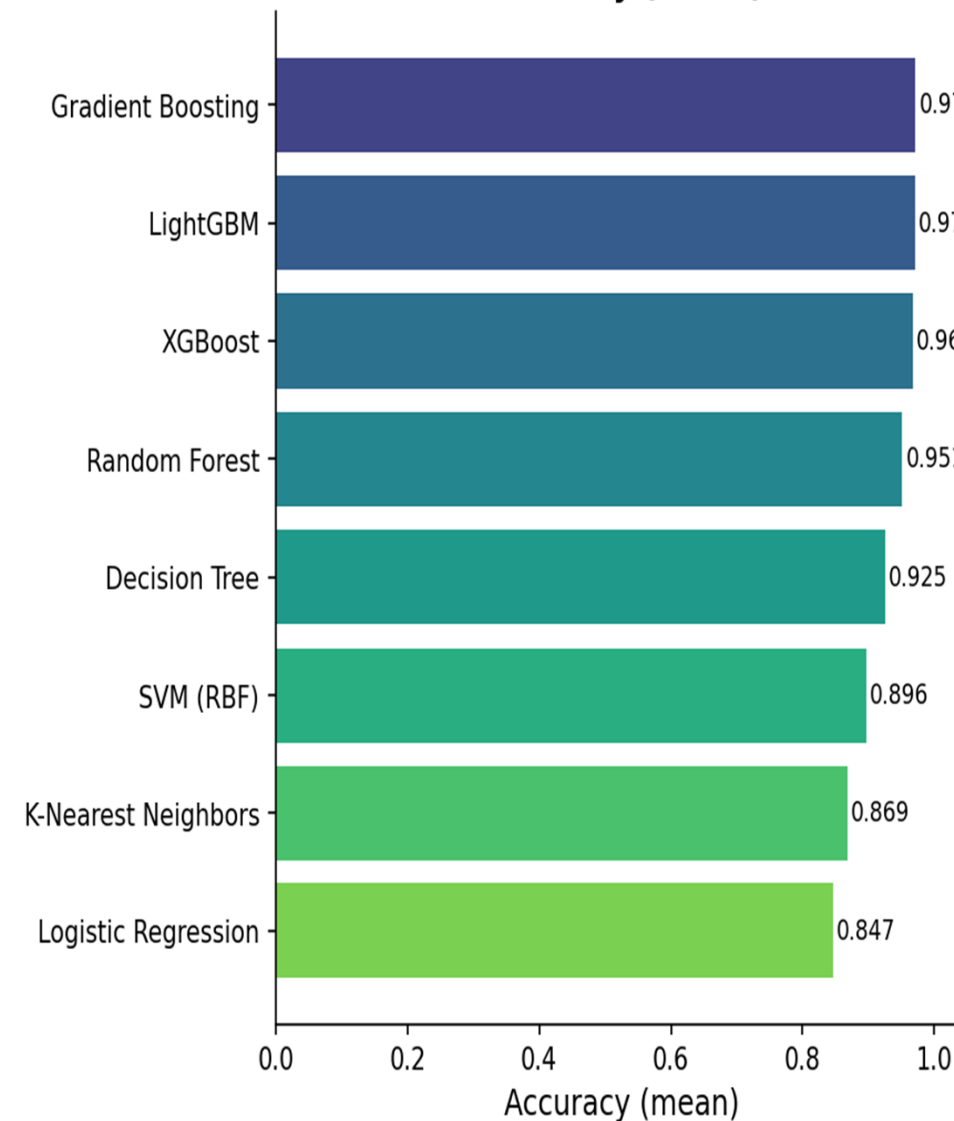
F1 (mean)



AUC (mean)



Accuracy (mean)

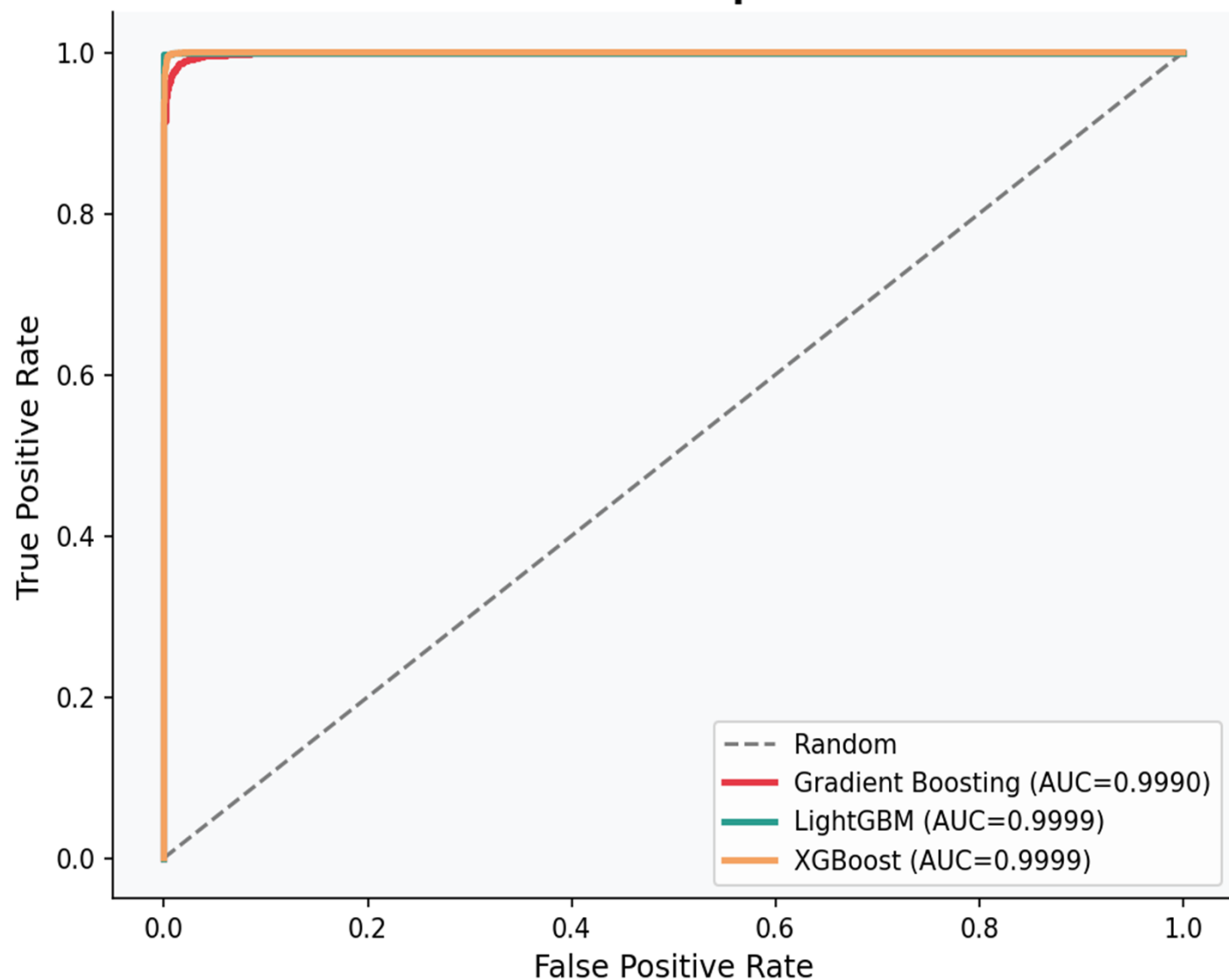




#1: Gradient Boosting

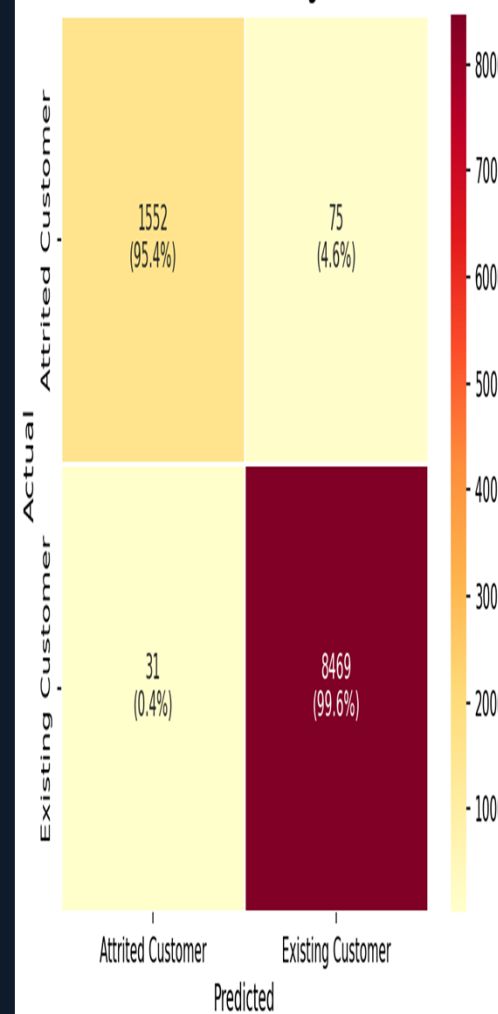
F1: 0.9830 | AUC: 0.9925 | Accuracy: 0.9714 | Precision: 0.9771 | Recall: 0.9891

ROC Curves — Top 3 Models

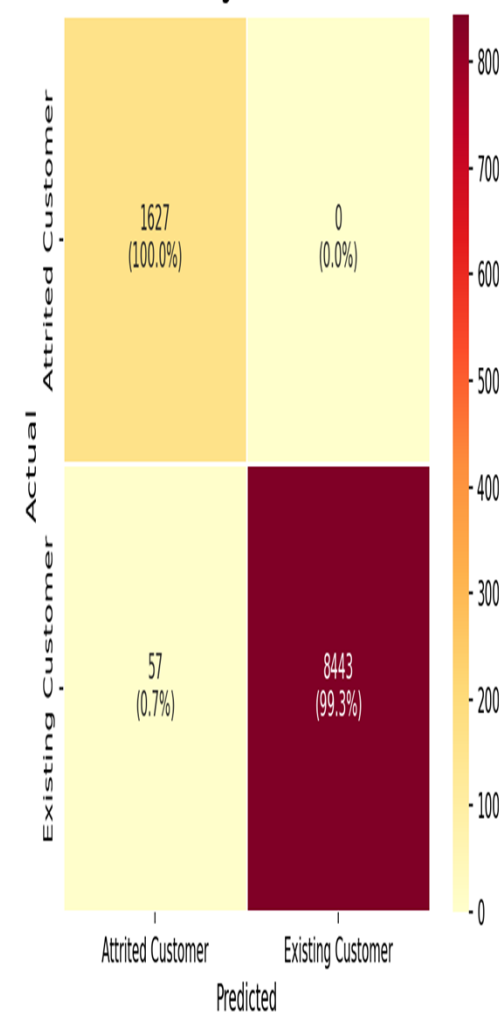


Confusion Matrices — Top 3 Models

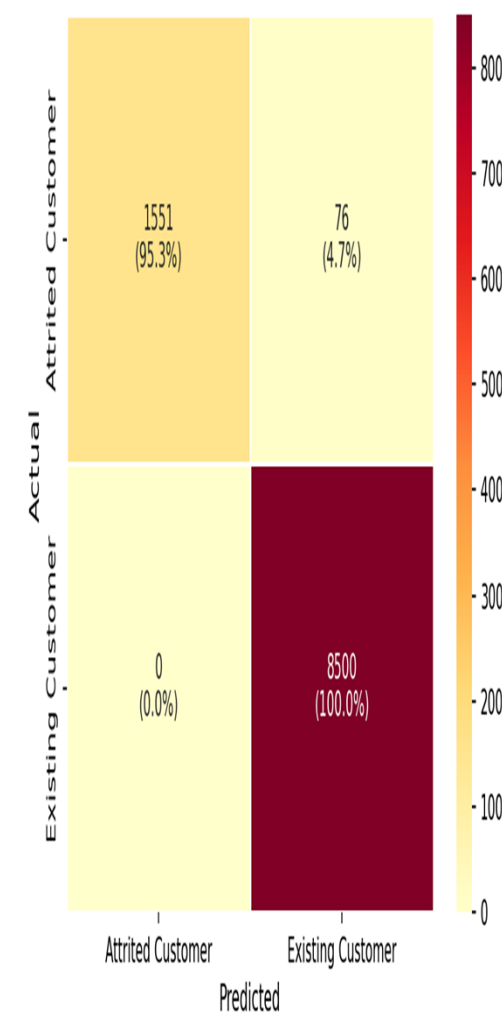
Gradient Boosting



LightGBM



XGBoost

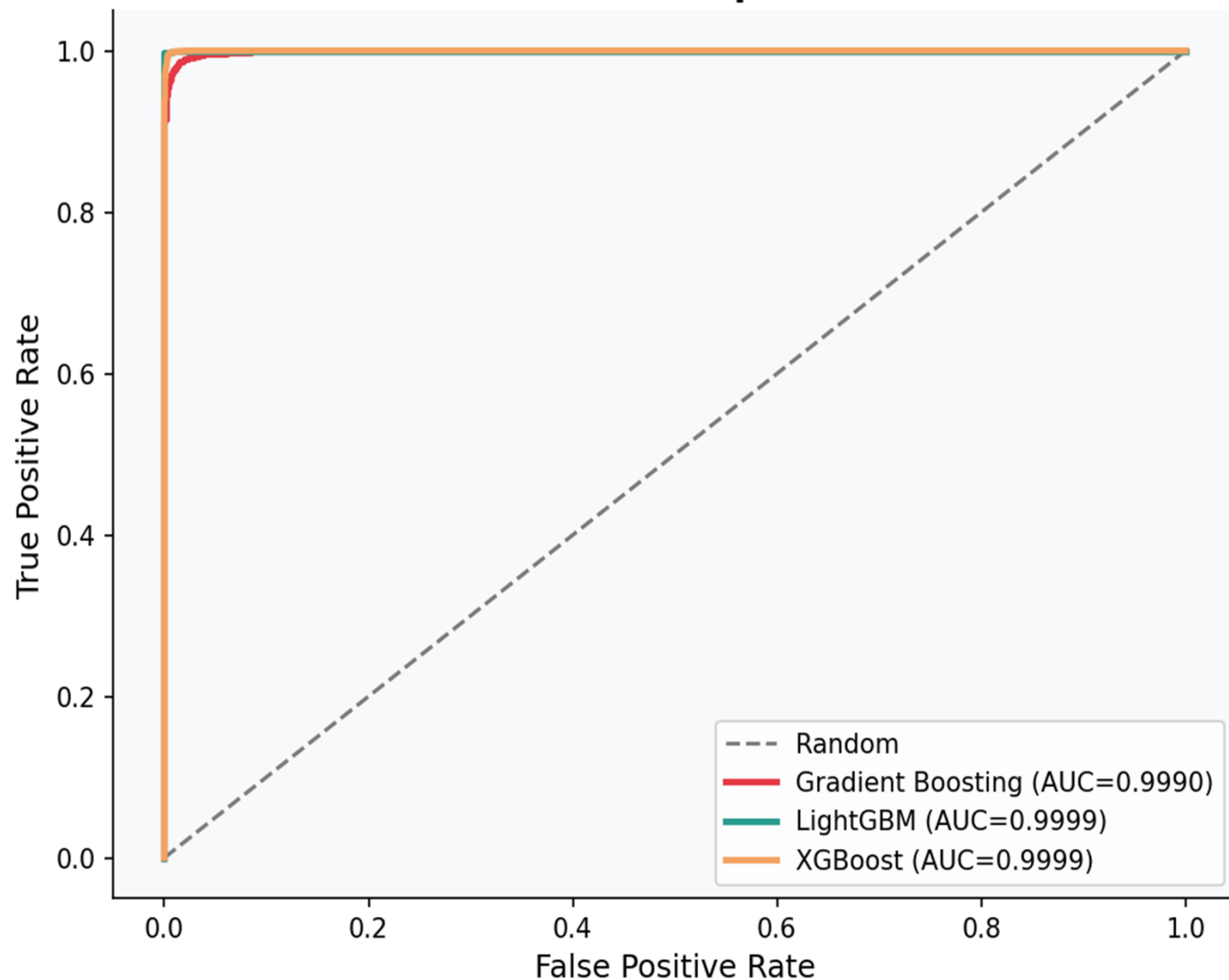




#2: LightGBM

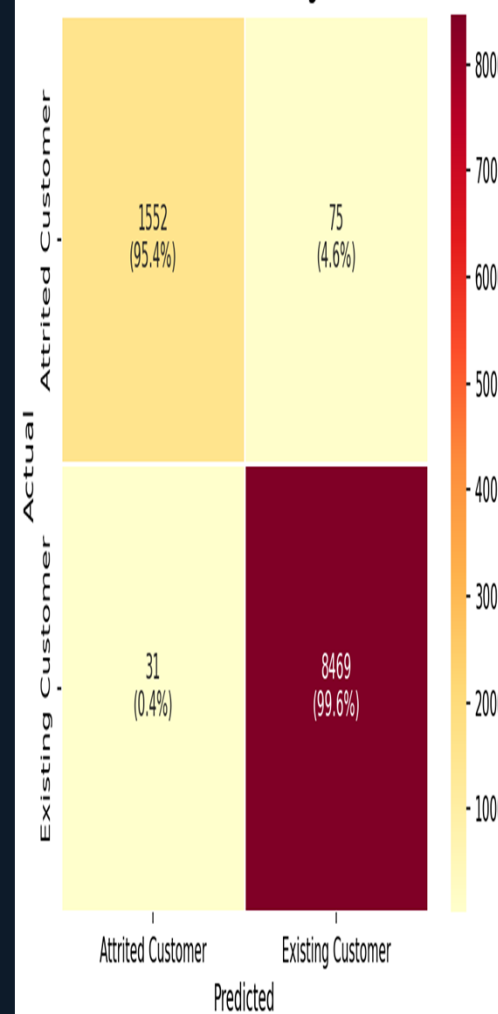
F1: 0.9824 | AUC: 0.9936 | Accuracy: 0.9706 | Precision: 0.9876 | Recall: 0.9772

ROC Curves — Top 3 Models

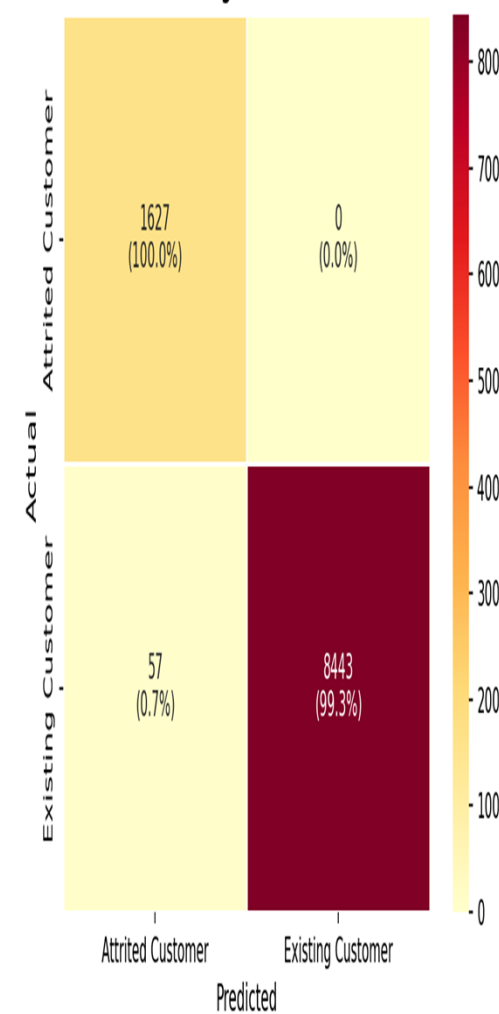


Confusion Matrices — Top 3 Models

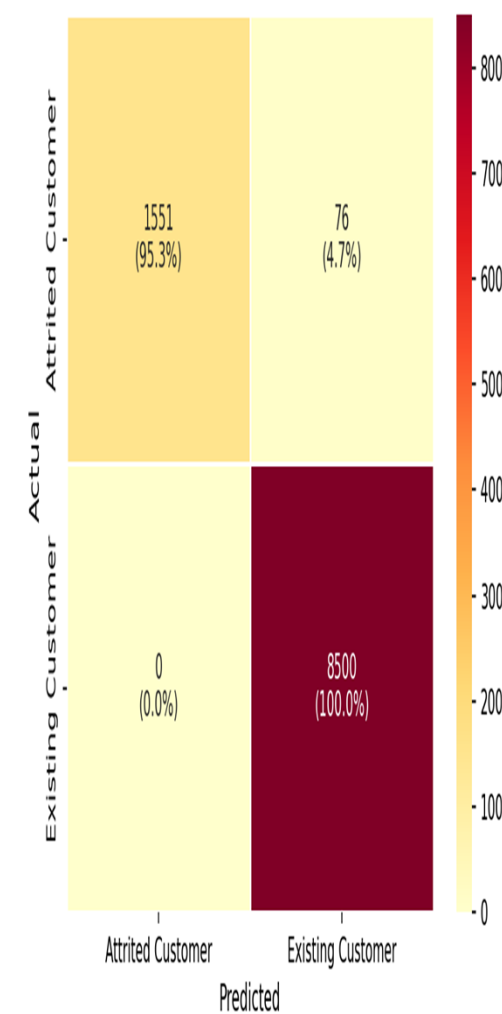
Gradient Boosting



LightGBM



XGBoost

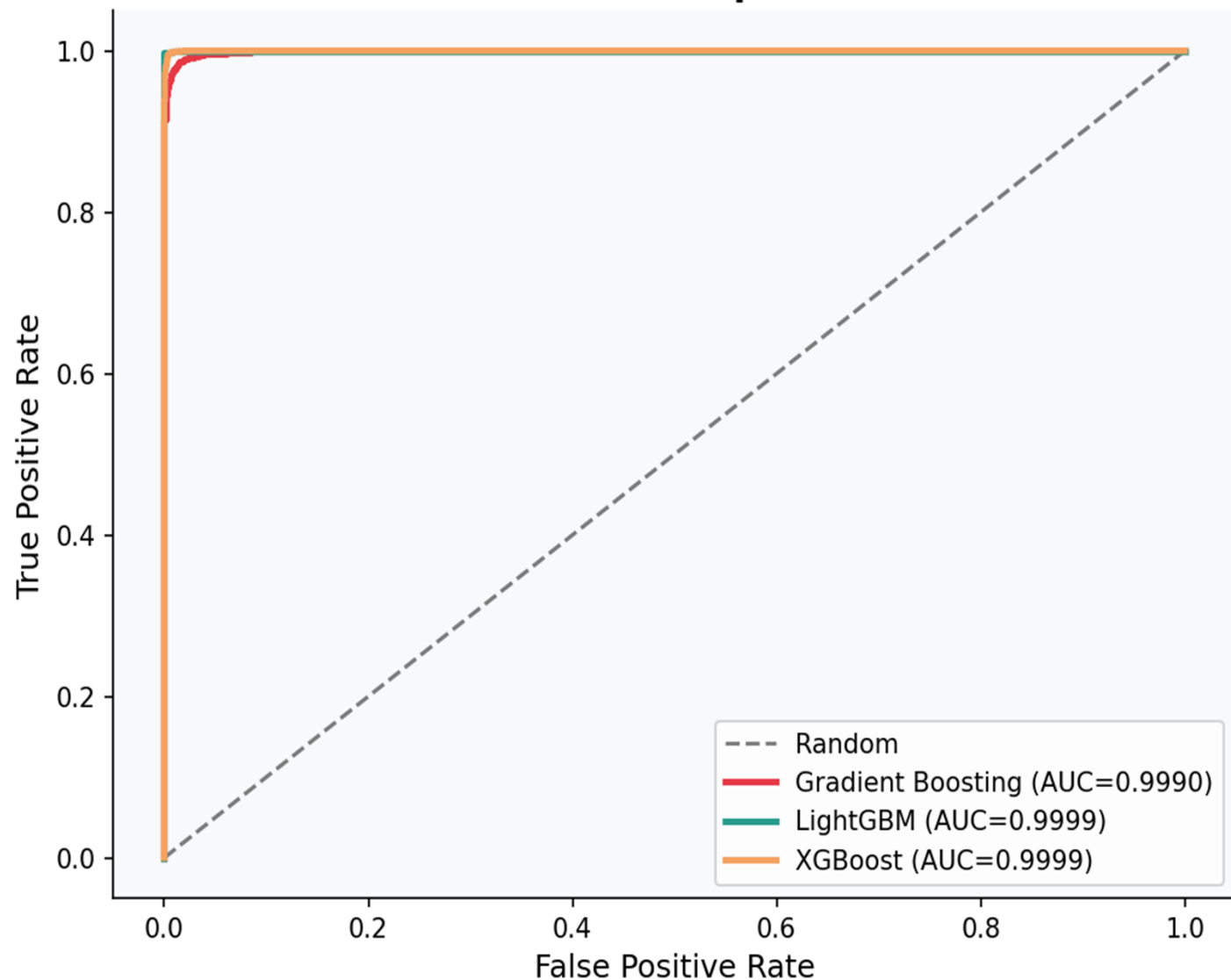




#3: XGBoost

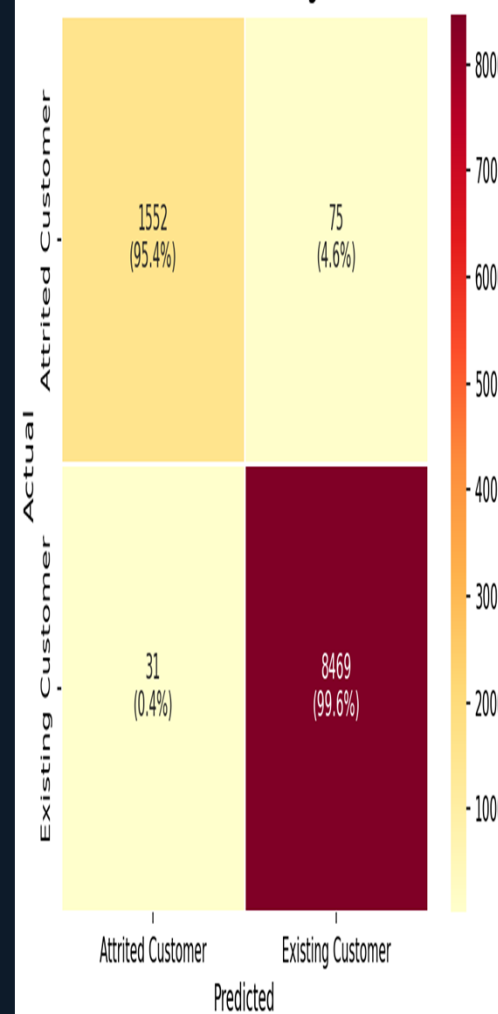
F1: 0.9809 | AUC: 0.9934 | Accuracy: 0.9675 | Precision: 0.9682 | Recall: 0.9940

ROC Curves — Top 3 Models

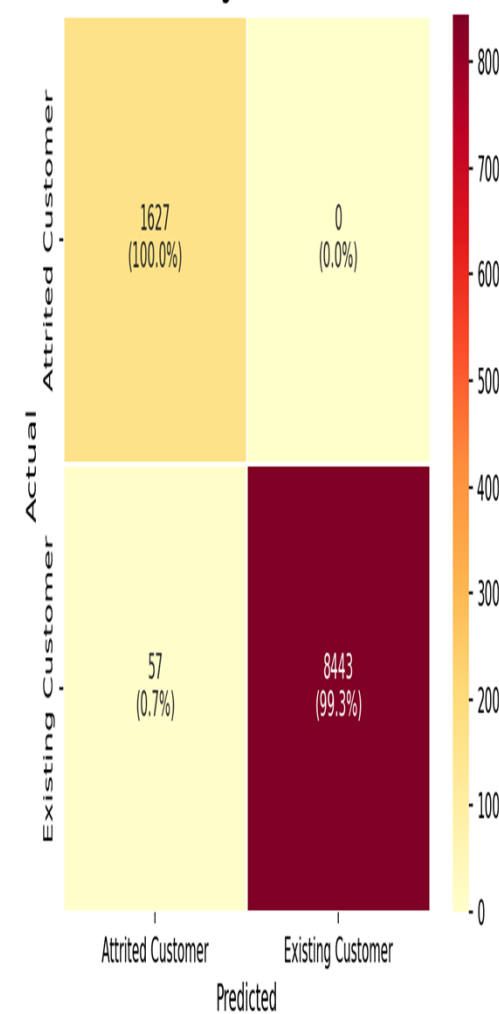


Confusion Matrices — Top 3 Models

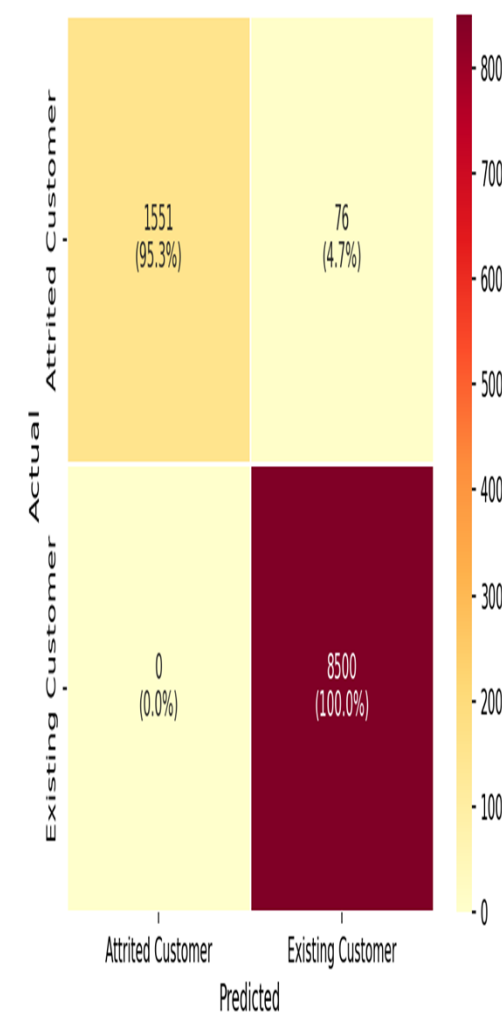
Gradient Boosting



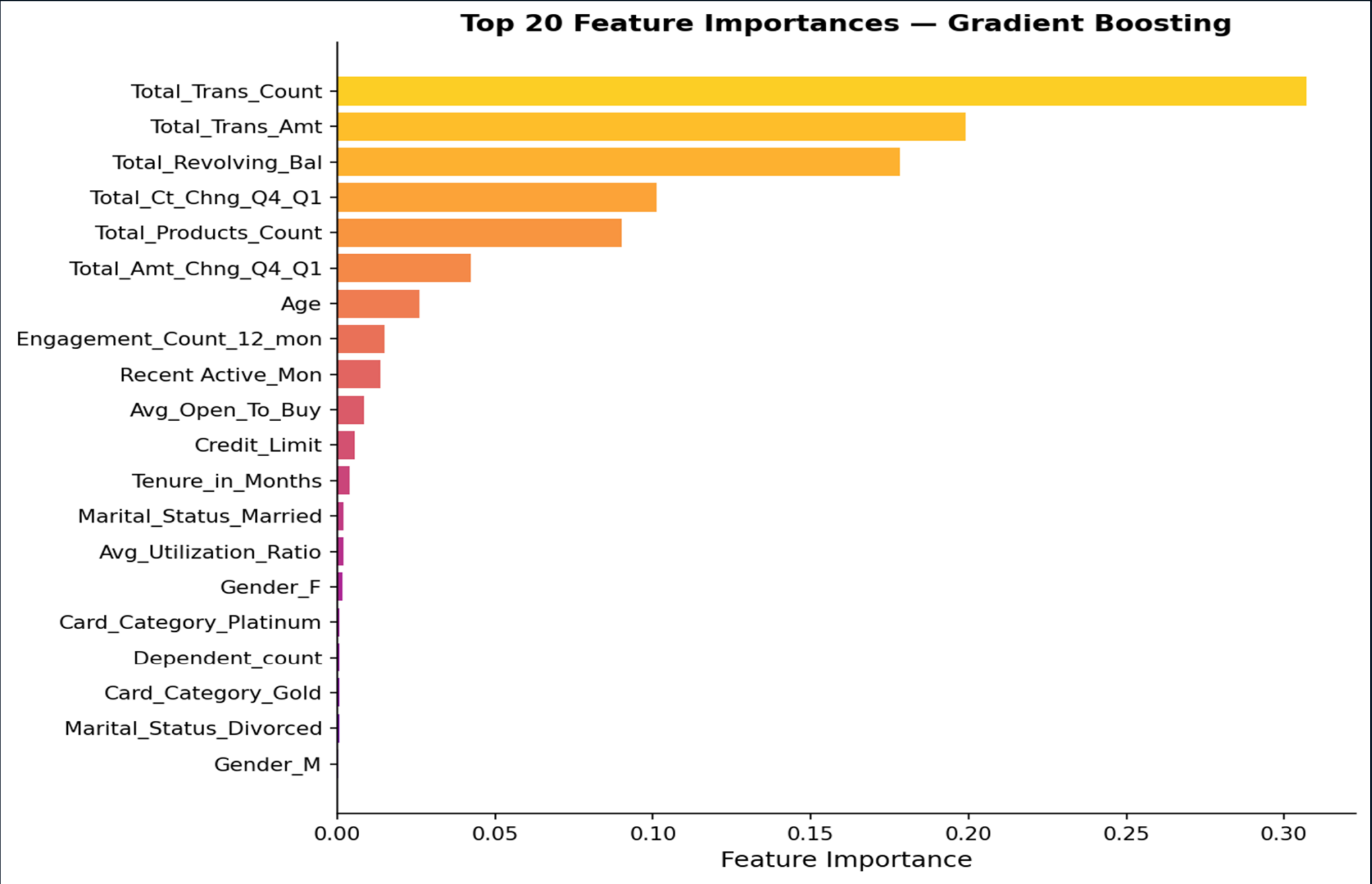
LightGBM



XGBoost



Feature Importance — Gradient Boosting





Hyperparameter Tuning — Top 3 Models

◆ Gradient Boosting

Best CV F1: 0.9830 | Best AUC: 0.9995

Best Params: {'learning_rate': 0.1, 'max_depth': 4, 'n_estimators': 200}

◆ LightGBM

Best CV F1: 0.9841 | Best AUC: 1.0000

Best Params: {'learning_rate': 0.2, 'max_depth': 5, 'n_estimators': 200}

◆ XGBoost

Best CV F1: 0.9819 | Best AUC: 1.0000

Best Params: {'learning_rate': 0.2, 'max_depth': 5, 'n_estimators': 200}



Key Findings & Recommendations

- ✓ Best Model: Gradient Boosting — highest F1-score in 5-fold cross-validation.
- ✓ Top 3 models (Gradient Boosting, LightGBM, XGBoost) all demonstrate strong churn prediction.
- ⚠ Class imbalance (16% churn) handled via `class_weight='balanced'` and `scale_pos_weight`.
- 📊 AUC > 0.9 confirms high discriminative power for all top-3 models.
- 🔧 Hyperparameter tuning (GridSearchCV) further improved F1 performance.
- 🚀 For production: deploy top model with threshold tuning to balance precision/recall.
- 📁 All results saved to `bank_results.csv` and `bank_tuning_results.csv`.